Basic Probability Theory

Chapter 6

Biostatistics and Data Analysis

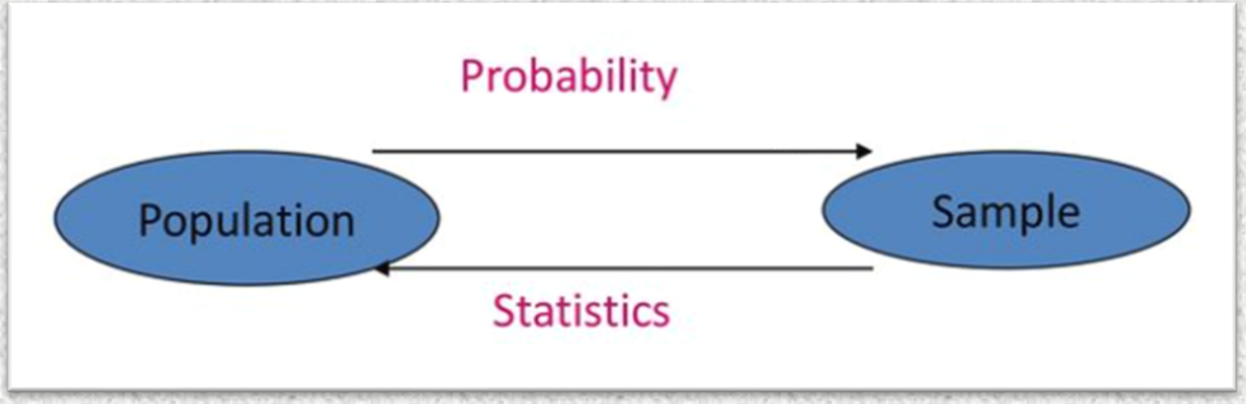
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- Multiplication Rule & Addition Rule
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INTRODUCTION TO PROBABILITY

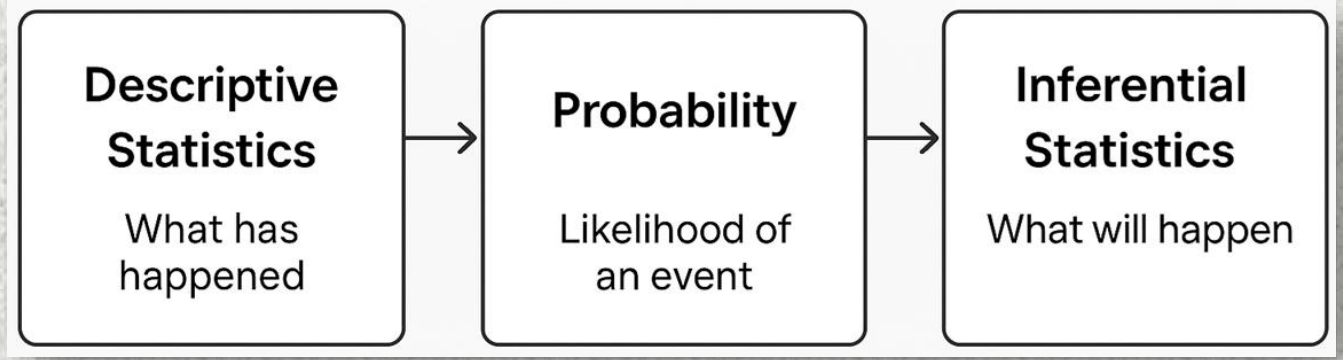
Why we study/learn Probability?

- Nothing in life is certain.
- In life whatever we do, we gauge it in terms of its successful outcomes, i.e
  - ✓ Outcome from business (Profit)
  - ✓ Weather (Chances of rain or sun)
- It provides a bridge between descriptive and inferential statistics.



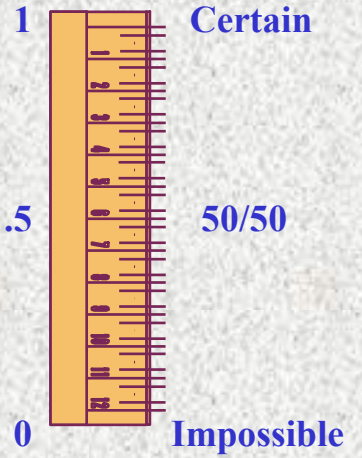
Why we study/learn Probability?

How bridge between descriptive and inferential statistics?



- **Descriptive (What happened) =** A shopkeeper sold more ice cream on sunny days than cloudy ones.
- **Probability (How likely) =** The forecast says there's a 70% chance of sun tomorrow, so, he will sale more ice cream.
- **Inferential (What will happen) =** He predicts high sales tomorrow.

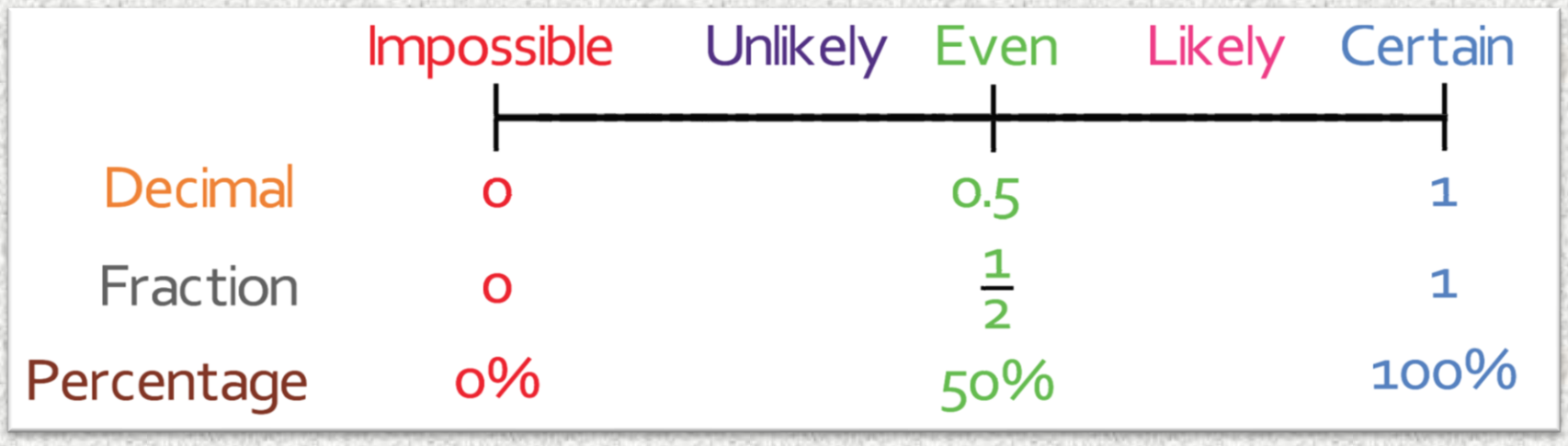
Probability



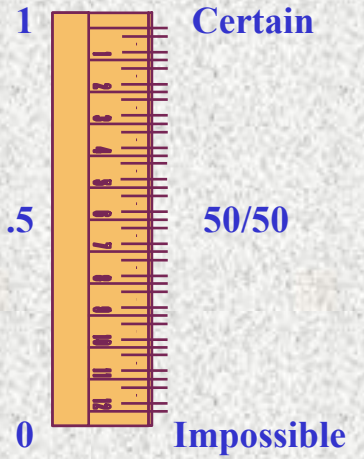
Fundamental concept of probability

"A Measure of Uncertainty"

- Probability is a measure of how likely an event is to occur.
- It tells us the relative frequency with which we can expect an event to occur.
- The greater the probability the more likely the event will occur.
- It can be written as a fraction, decimal, percent, or ratio.

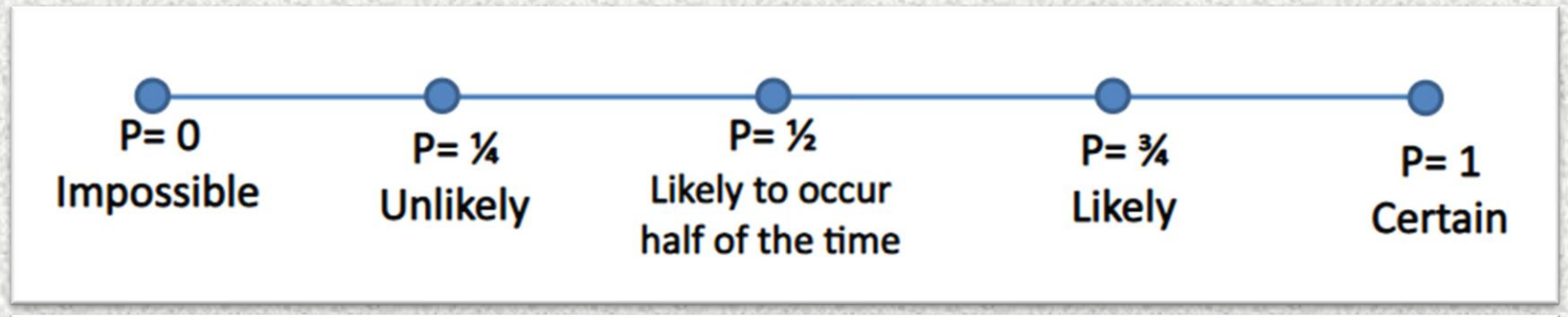


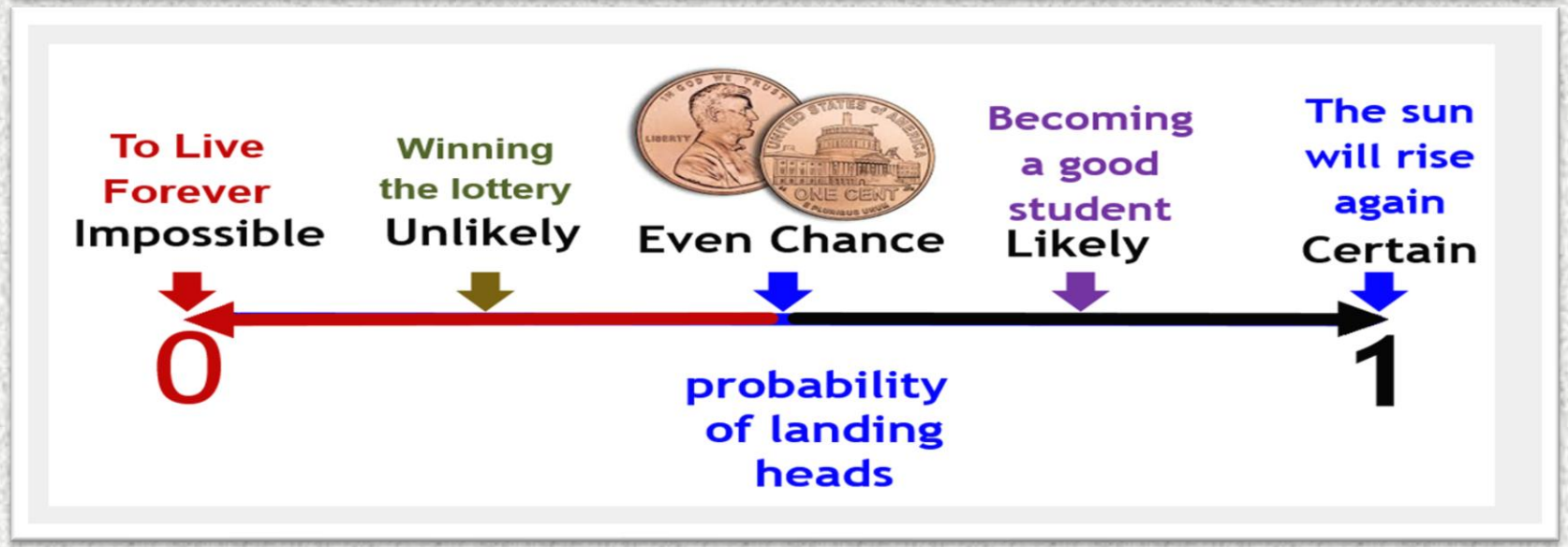
Probability



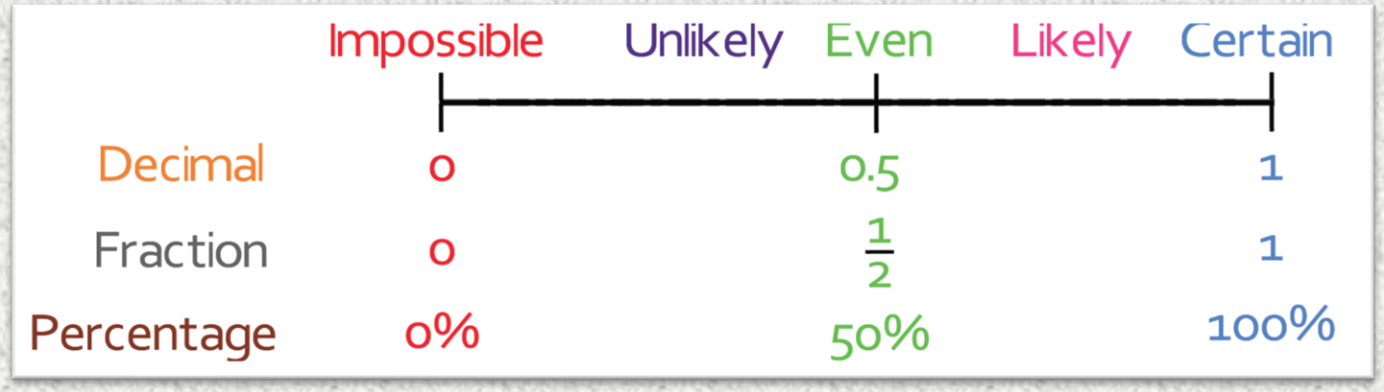
Fundamental concept of probability

- Probability quantifies the likelihood of an event occurring, ranging from 0 (impossible) to 1 (certain).





Probability

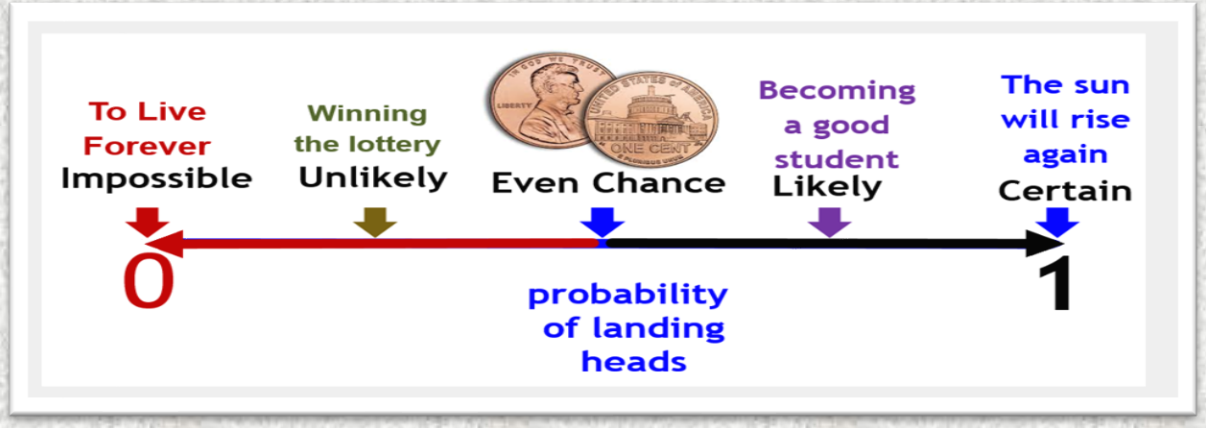


Understanding Probability: Decimal, Fraction & Percentage

- The probability of any event ***E***, written as **"*P*(*E*)"**, always lies between "0 & 1" and written like: $0 \leq P(E) \leq 1$
- If an event **can never happen**, then **P(E) = 0** or **0%**
- If an event is **equally likely to happen or not**, then **P(E) = 0.5, ½ , 50%**
- If an event is **certain to happen**, then **P(E) = 1** or **100%**.

Probability

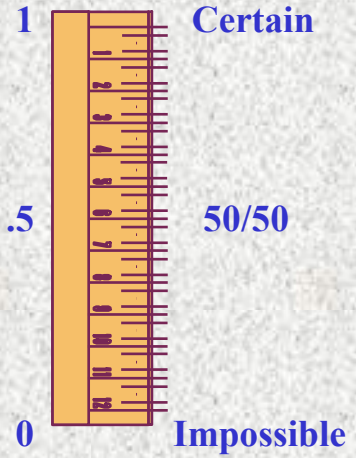




Understanding Probability: Impossible, Unlikely, Even Chances, Likely, and Certain

- When a meteorologist says there is a **0% chance of rain**, it means it is **impossible** for it to rain.
- If the chance of rain rises **to 30%**, it means it is **unlikely** to rain.
- If the chance of rain is **50%**, it means it is **equally likely** to rain or not to rain.
- If the chance of rain **rises to 80%**, it means it is **likely** to rain.
- If the chance of rain reaches **100%**, it means it is **certain** to rain.

Probability



Fundamental concept of probability

- **Examples**

Weather:

1. 30% chance of sunshine tomorrow
2. 80% chance of thunderstorms tonight
3. 50% chance of fog in the morning

Sports:

1. 70% chance of Team A winning the game
2. 25% chance of a draw in the match
3. 40% chance of Player X scoring a goal

Finance:

1. 20% chance of stock prices increasing tomorrow
2. 60% chance of market growth this quarter
3. 15% chance of recession next year

Health:

1. 80% chance of recovery from surgery
2. 40% chance of success with medication
3. 10% chance of complications from treatment

Travel:

1. 50% chance of flight delays
2. 30% chance of lost luggage
3. 80% chance of clear roads

Basic Concepts

- Experiment
- Event
- Sample space



Basic Concepts

Experiment and Event



Experiment:

- A **process** that produces an **observation or measurement**.
  - 👉 *Example:* Tossing a coin or rolling a die is an experiment.

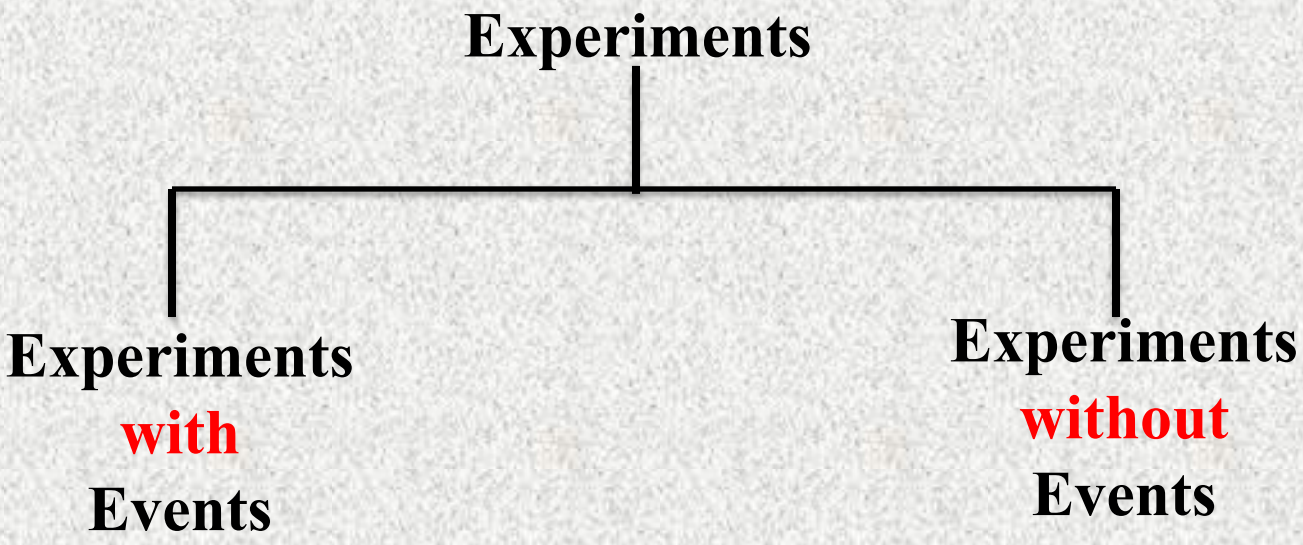
Event:

- A **specific outcome** of an experiment, usually denoted by a **capital letter** such as **A** or **B**.
  - 👉 *Example:* In a coin toss, **Event A** could be *getting heads*.

    In a die roll, **Event B** could be *rolling a 4*.

Basic Concepts

Experiments and Events



Basic Concepts

Experiments and Events

Example of Experiments with Events



1. **Experiment:** Coin Toss
   - ✓ Event A: Head
   - ✓ Event B: Tail

2. **Experiment:** Medical Test
   - ✓ Event A: Positive result
   - ✓ Event B: Negative result

3. **Experiment:** Weather Forecast
   - ✓ Event A: Rain
   - ✓ Event B: No rain



Basic Concepts



Experiments and Events

Example of Experiments without Events

1. **Experiment:** Coin Toss
   - ✓ No Event: The coin lands on its edge (virtually impossible)

2. **Experiment:** Medical Test
   - ✓ No Event: Test results are inconclusive or invalid, so no clear outcome is obtained.

3. **Experiment:** Weather Forecast
   - ✓ No Event: Weather cannot be determined (e.g., instrumentation failure)

Basic Concepts

Experiments and Events



Dependent events Vs Independent events:

- **Dependent events:**
  - Two events are said to be **dependent** if the occurrence of the **first event** affects the **probability of the second event**.
  - **Example:**

    Event A: It rains today

    Event B: It rains tomorrow
  - If it rains today, the **likelihood of rain tomorrow increases**, showing that the two events are **dependent**.

Basic Concepts

Experiments and Events

Dependent events Vs Independent events:

- **Independent events:**
  - Two events are said to be **independent** when the occurrence of the **first event** does **not affect** the probability of the **second event**.
  - **Example:**

    Event A: Coin 1 lands on heads

    Event B: Coin 2 lands on heads



  - The outcome of **Coin 1** does **not influence** the outcome of **Coin 2**, so these events are **independent**.

Basic Concepts

Experiments and Events

Dependent events Vs Independent events:

Feature	Dependent Events	Independent Events
Definition	The occurrence of the first event **affects** the probability of the second event.	The occurrence of the first event **does not affect** the probability of the second event.
Relationship	The two events are **connected**.	The two events are **unrelated**.
Example	**Event A**: Rain today **Event B**: Rain tomorrow Rain today increases the chance of rain tomorrow.	**Event A**: Coin 1 lands on heads **Event B**: Coin 2 lands on heads The outcome of Coin 1 does not affect Coin 2.
Result Influence	One event **changes** the likelihood of the other.	One event **has no effect** on the other.

Basic Concepts

Simple Event vs Sample Space

Simple Event:

Definition:

- A **simple event** is a **single possible outcome** of an experiment.

Key Points:

- Represents **one specific result**.
- Each simple event is a **part of the sample space.**

Basic Concepts

Simple Event vs Sample Space

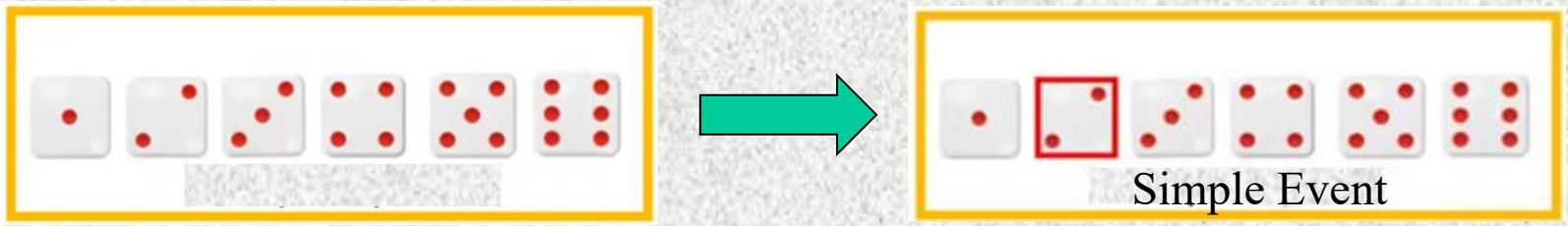
Simple Event:

Examples:-

- **Coin Toss:** H (Heads) OR T (Tails)



- **Roll of a Die:** 1 OR 2 OR 3 OR 4 OR 5 OR 6



Basic Concepts

Simple Event vs Sample Space

Sample Space:

Definition:

- The **sample space** is the **set of all possible outcomes** of an experiment.

Key Points:

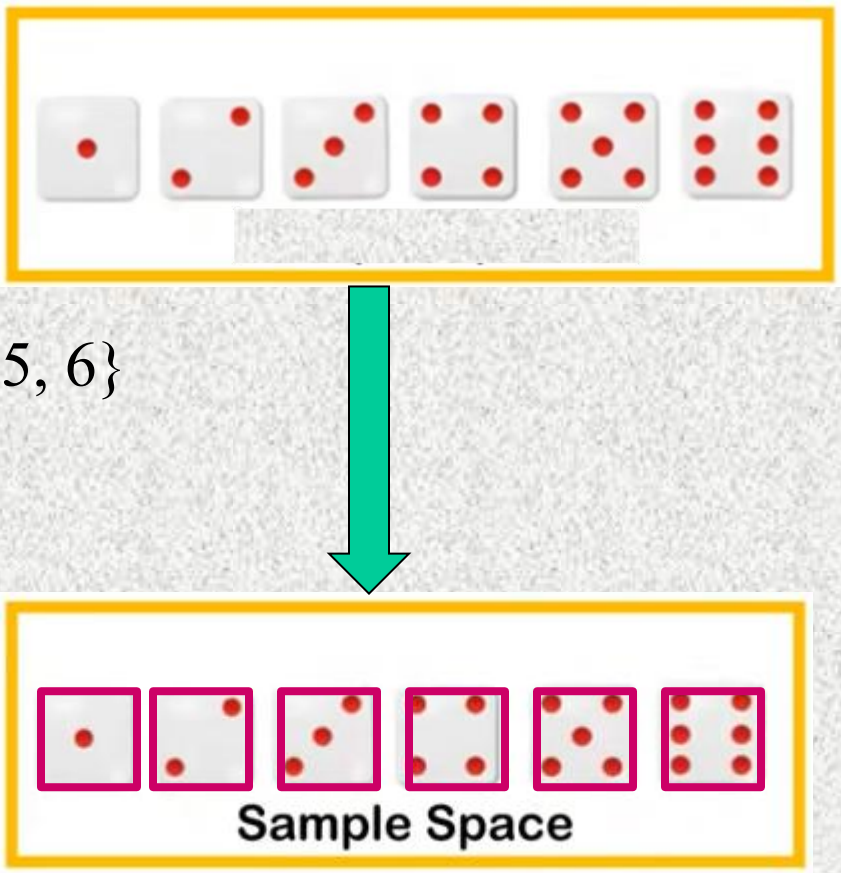
- Includes **every possible result**.
- Simple events **make up** the sample space.

Basic Concepts

Simple Event vs Sample Space

Sample Space:

- Examples:-
  - Coin Toss: {H, T}
  - Roll of a Die: {1, 2, 3, 4, 5, 6}
  - Medical Test: {+, -}

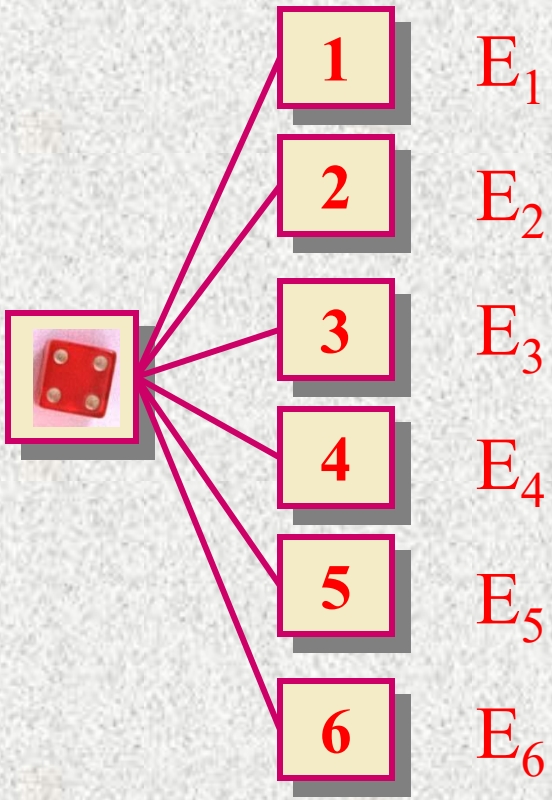


Example: Experiment of "The Die Toss"



- **Simple event:**

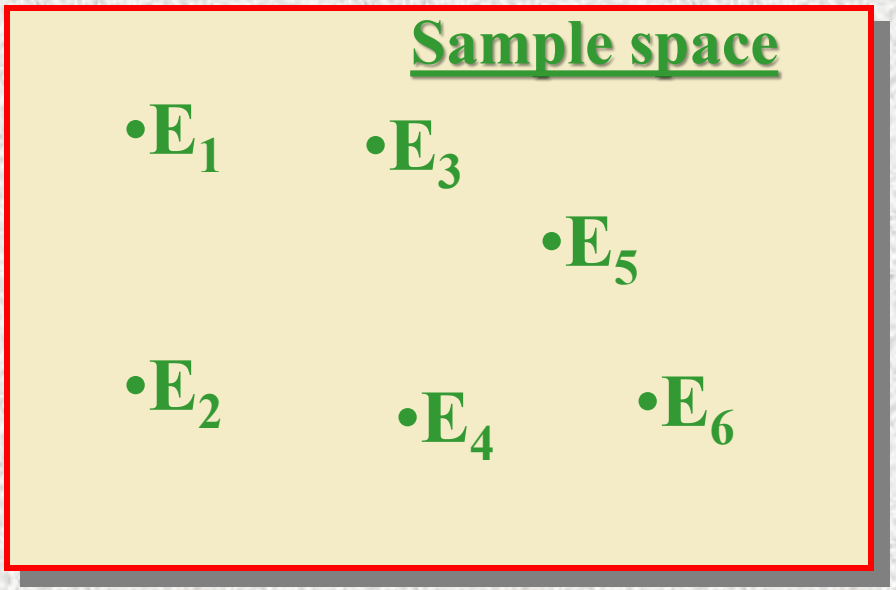
A **simple event "E"** is any single outcome from a probability experiment.



Sample space:

The **sample space "*S*"** of a probability experiment is the collection of all possible simple events.

$$S = \{E_1, E_2, E_3, E_4, E_5, E_6\}$$



TYPES OF PROBABILITY

Types of Probability

Different types of probability are enlisted below:

1. **Theoretical Probability**
2. **Experimental Probability**
3. **Subjective Probability**
4. **Objective Probability**

Types of Probability

1. Theoretical Probability

- Theoretical probability is used when each outcome in a sample space has equal chance of occurring.

- Equation for "**Theoretical Probability**" is given below:

$$\mathbf{P(E) = \frac{Number\ of\ Event\ Outcomes}{Total\ Number\ of\ Possible\ Outcomes\ in\ S}}$$

- **Example:**
  When tossing a fair coin,

$$P(Head) = \frac{1}{2}$$



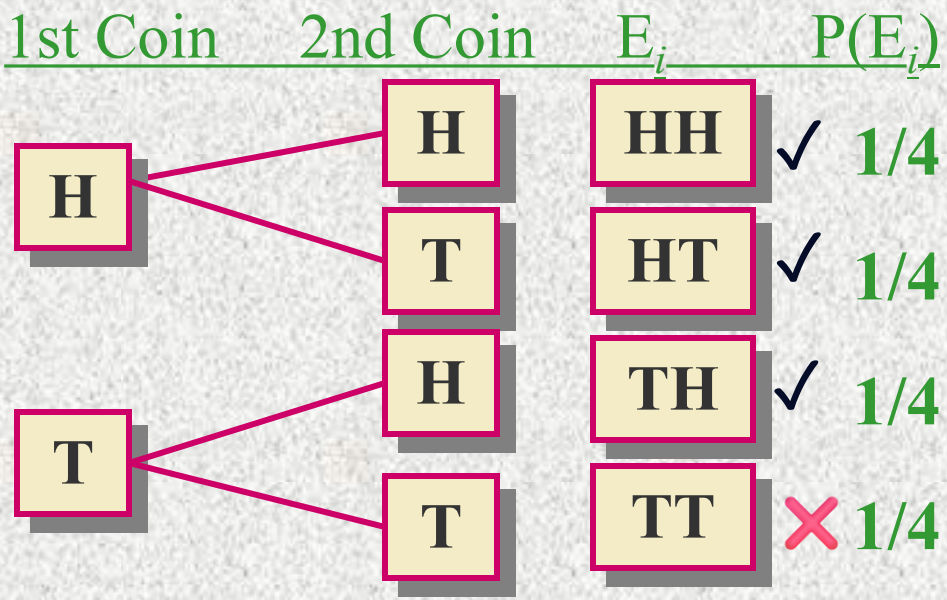
Types of Probability

1. Theoretical Probability (Example 1)

- Toss a fair coin twice. What is the probability of observing at least one head?

$$P(E) = \frac{\text{Number of Event Outcomes}}{\text{Total Number of Possible Outcomes in S}}$$





P(at least 1 head)

$= P(E_1) + P(E_2) + P(E_3)$

$= 1/4 + 1/4 + 1/4$

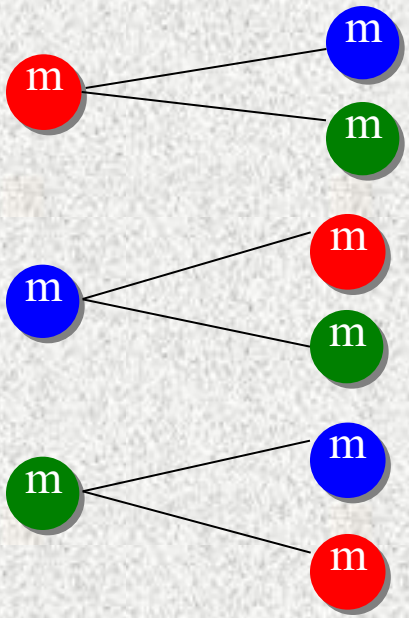
$= 3/4$

Example 2

- A bowl contains three M&Ms®, one red, one blue and one green. A child selects two M&Ms at random. What is the probability that at least one is red?

$$P(E) = \frac{\text{Number of Event Outcomes}}{\text{Total Number of Possible Outcomes in S}}$$





1st M&M	2nd M&M	E_i		$P(E_i)$
m	m	RB	✓	1/6
m	m	RG	✓	1/6
m	m	BR	✓	1/6
m	m	BG	✗	1/6
m	m	GB	✗	1/6
m	m	GR	✓	1/6

P(at least 1 red)

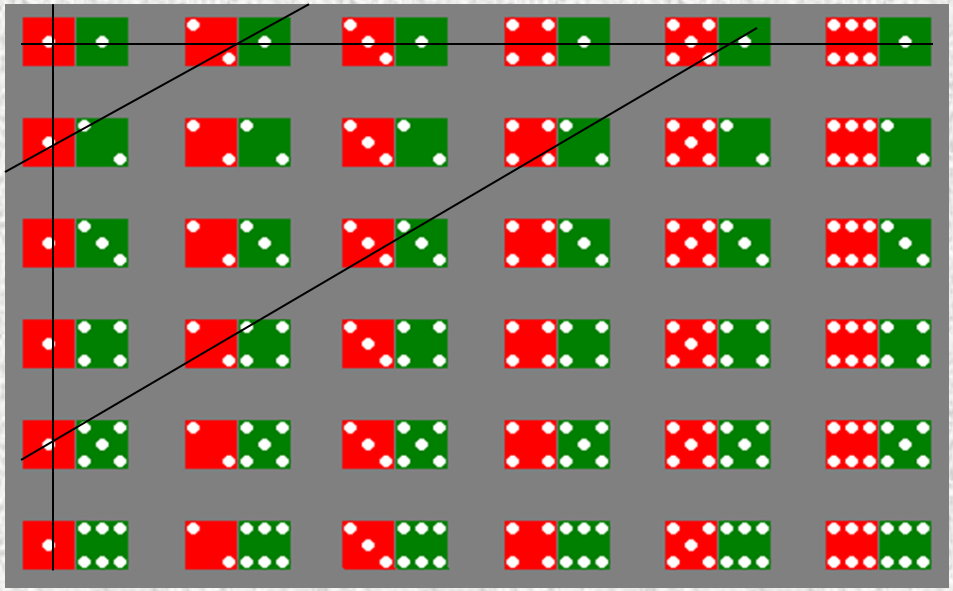
$= P(E_1) + P(E_2) + P(E_3) + P(E_6)$

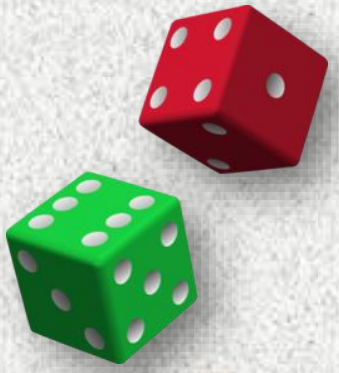
$= 4/6$

$= 2/3$

Example 3

- The sample space of throwing a pair of dice is





$$P(E) = \frac{\text{Number of Event Outcomes}}{\text{Total Number of Possible Outcomes in S}}$$

Event	Simple events	Probability
Dice add to 3	(1,2),(2,1)	2/36
Dice add to 6	(1,5),(2,4),(3,3), (4,2),(5,1)	5/36
Red die show 1	(1,1),(1,2),(1,3), (1,4),(1,5),(1,6)	6/36
Green die show 1	(1,1),(2,1),(3,1), (4,1),(5,1),(6,1)	6/36

Types of Probability

2. Experimental Probability

- Experimental probability is based upon observations obtained from probability experiments.

- Equation for "**Experimental Probability**" is given below:

$$\mathbf{P(E) = \frac{Number\ of\ times\ an\ Event\ Occurs}{Total\ Number\ of\ Experiments}}$$

- **Example:**

  If a coin is tossed 10 times and shows heads 6 times,

$$P(Head) = \frac{6}{10} = 0.6$$

Types of Probability





2. Experimental Probability (Example 1)

- **Fred rolled Dice 20 times. The results of his experiment are given below:**

Number	1	2	3	4	5	6
Times Rolled	3	4	2	6	1	4

- What is the experimental probability of rolling an even number?

$$P(E) = \frac{\text{Number of times an Event Occurs}}{\text{Total Number of Experiments}}$$

Number	1	2	3	4	5	6
Times Rolled	3	4	2	6	1	4

$$P(E) = \frac{4+6+4}{20} \qquad P(E) = \frac{14}{20} \qquad P(E) = \frac{7}{10}$$

Example 2

Finding Probability from a Frequency Chart

- The following chart that gives the marks of a total of 1200 students in an examination.

Marks	No. of Students
100-120	65
120-140	140
140-160	500
160-180	320
180-200	175

If we select a student then find the probability that the student has marks:

i) Above 180

ii) Between 140 and 180

iii) Below 120

Example 2

Marks	No. of Students
100-120	65
120-140	140
140-160	500
160-180	320
180-200	175

Solution:

i) Above 180

- 175 students have marks above **180.**
- P (marks above 180) = 175/1200 = 0.15

ii) Between 140 and 180

- Number of students getting marks between 140 and 180 are **500 + 320 = 820.**
- Hence P (marks between 140 and 180) = 820//1200 = 0.68

iii) Below 120

- Number of students getting marks below 120 are **65**.
- Hence, P (marks under 120) = 65/1200 = 0.05

Theoretical Probability
Vs
Experimental Probability

How can you tell which is experimental and which is theoretical probability?

Experimental:

You tossed a coin 10 times and recorded a head 3 times, a tail 7 times.

$$P(\text{head}) = \frac{3}{10}$$

$$P(\text{tail}) = \frac{7}{10}$$

Theoretical:

Toss a coin and getting a head or a tail is 1 out of 2 sides.

$$P(\text{head}) = \frac{1}{2}$$

$$P(\text{tail}) = \frac{1}{2}$$

Theoretical Probability
Vs
Experimental Probability

Experimental probability	Theoretical probability
6 appears 65 times. $P(\text{roll a } 6) = \frac{65}{100}$  Roll a die 100 times.	6 appears. $P(\text{roll a } 6) = \frac{1}{6}$  Roll a die once.

$$P(E) = \frac{\text{Number of times an Event Occurs}}{\text{Total Number of Experiments}}$$

$$P(E) = \frac{\text{Number of Event Outcomes}}{\text{Total Number of Possible Outcomes in S}}$$

Types of Probability

3. Subjective Probability

- Based on personal beliefs and opinions.
- Therefore, there is no formula to calculate it.
- Usually found by consulting an expert.
- It is a probability measure resulting from intuition, educated guesses, and estimates.
- Used in decision-making under uncertainty.



Types of Probability

3. Subjective Probability

Example

- Which team will win the football match?
- A. **England**
- B. Africa

England **Africa**



- Assuming, I have some knowledge of football, and my answer is “England”. Here’s my subjective probability assessment:
  - ✓ England has a strong football tradition, infrastructure, and recent good performances in international tournaments.
  - ✓ England has talented individual players.
  - ✓ England teamwork and coordination is excellent.
  - ✓ England’s recent international matches suggest that England's defense and attacking prowess is solid.

Types of Probability

4. Objective Probability

- Objective probability is a probability assessment based on objective data, measurements, and observations.
- It's independent of personal opinions, biases, and emotions.
- Used in Health & Medicine, finance, and engineering.

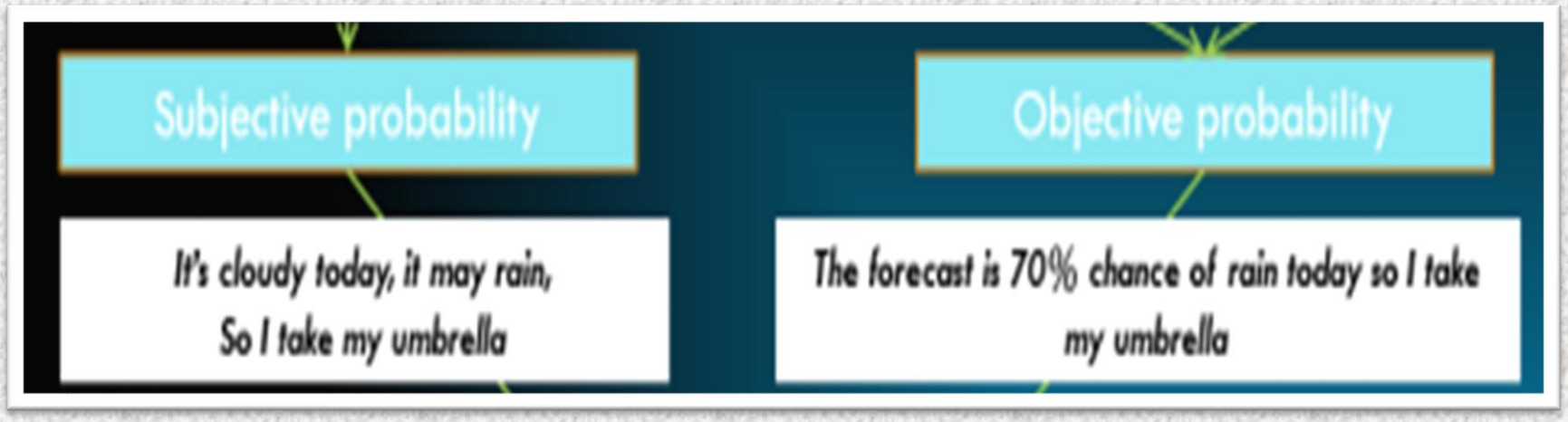
Types of Probability

4. Objective Probability

Examples

- **Health & Medicine**
  - ✓ Vaccine effectiveness: P(Vaccine prevents infection) = 0.9 (based on clinical trials)
- **Finance**
  - ✓ Currency exchange rate: P(USD/EUR > 1) = 0.6 (based on market trends)
- **Engineering**
  - ✓ Material strength: P(Failure under stress) = 0.1 (based on material testing)

Subjective Probability
Vs
Objective Probability



- **Event Relations**
  - ✓Union
  - ✓Intersection
  - ✓Conditional

Symbols Used In Event Relations

OR	$\cup$	UNION
AND	$\cap$	INTERSECTION
GIVEN	/	CONDITIONAL

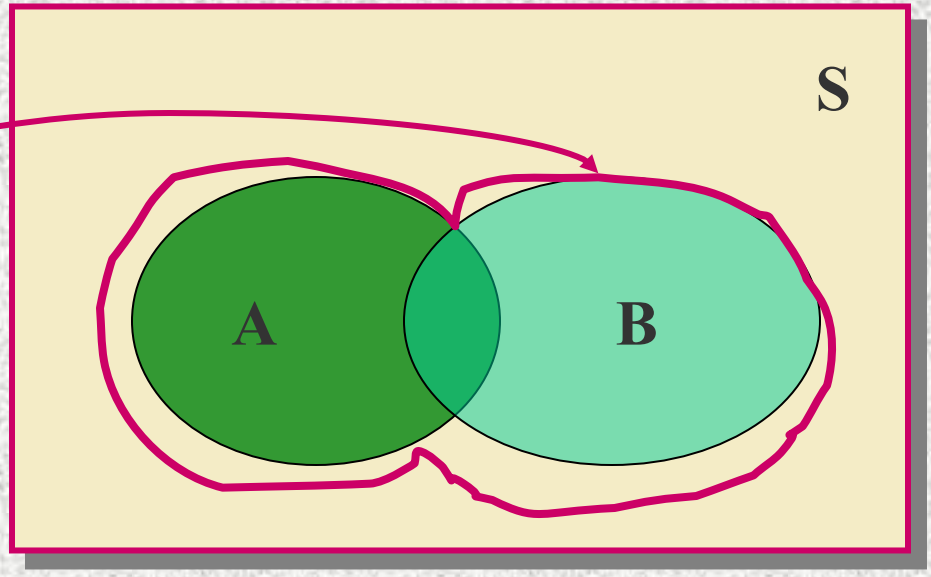
Event Relations

OR	$\cup$	UNION
AND	$\cap$	INTERSECTION
GIVEN	/	CONDITIONAL

Union of Events

- The union of events A and B, consists of all events that are in A or in B or in both A and B.
- We write "**A ∪ B**"

$A \cup B$



Event Relations

OR	$\cup$	UNION
AND	$\cap$	INTERSECTION
GIVEN	/	CONDITIONAL

Union of Events

Example:

Let event **A** = getting an even number when rolling a die = {2, 4, 6}

Let event **B** = getting a number greater than 3 = {4, 5, 6}

Then,

A ∪ B = {2, 4, 5, 6}

👉 This means the result is in **A**, **B**, or both.



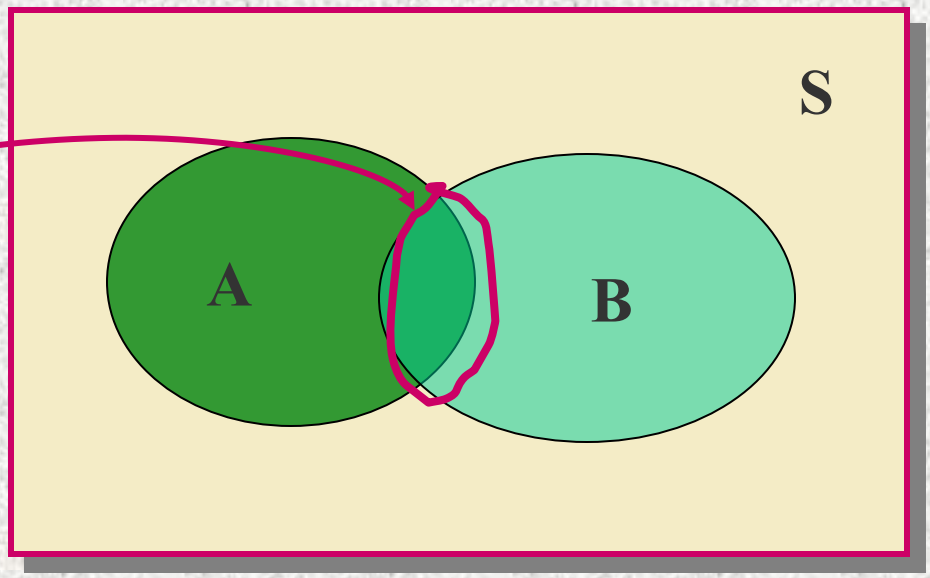
Event Relations

OR	$\cup$	UNION
AND	$\cap$	INTERSECTION
GIVEN	/	CONDITIONAL

Intersection of Events

- The **intersection of events A and B** includes all outcomes that are **common to both A and B**.
- We write "**A ∩ B**"

$A \cap B$



Event Relations

OR	$\cup$	UNION
AND	$\cap$	INTERSECTION
GIVEN	/	CONDITIONAL

Intersection of Events

Example:

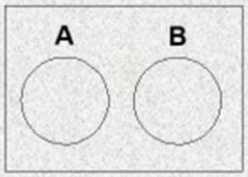
Let **A** = {2, 4, 6}

B = {4, 5, 6}

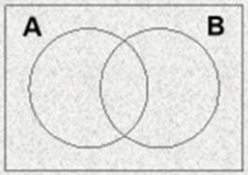
Then, **A** $\cap$ **B** = {4, 6}

👉 These are the outcomes that occur in **both events**.

Event Relations



A and B are M.E.



A and B are not M.E.

Intersection of Events

Mutually Exclusive

- Two events A and B are mutually exclusive if and only if:

$$P(A \cap B) = 0$$

- This means that the probability of both events occurring simultaneously is zero.

- In a Venn diagram this means that event A is disjoint from event B.
- Meaning they:
  - 1. Do not overlap
  - 2. Have no common elements
  - 3. Are separate and distinct

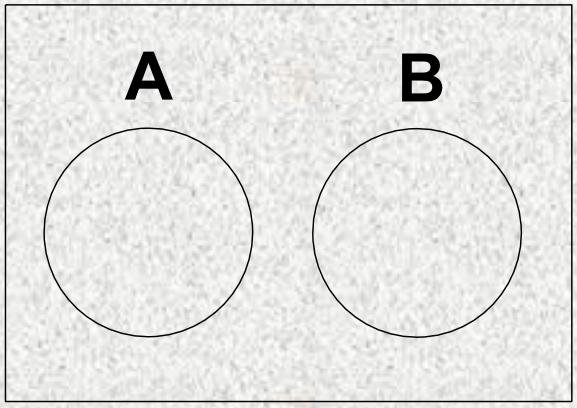
Event Relations

Intersection of Events

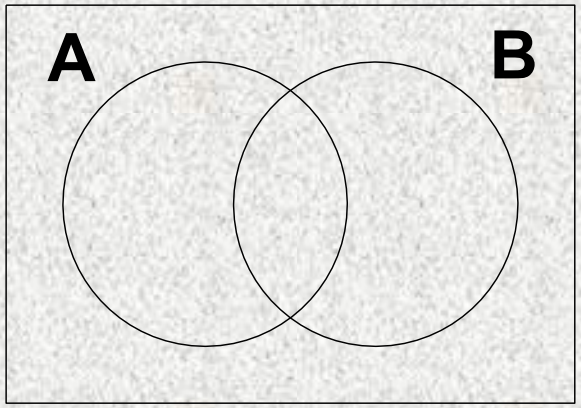
Mutually Exclusive

VENN DIAGRAM

- Event A [Circle A]
- Event B [Circle B]



A and B are M.E.



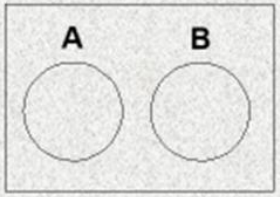
A and B are not M.E.

Event Relations

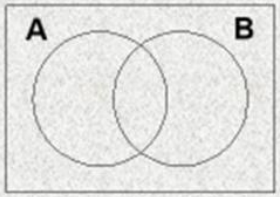
Intersection of Events

Mutually Exclusive

Example:



A and B are M.E.



A and B are not M.E.

When rolling a die:

- **A** = getting an even number = {2, 4, 6}
- **B** = getting an odd number = {1, 3, 5}

$$P(A \cap B) = 0$$

Since no outcome is common, **A** and **B** are mutually exclusive.

Multiplication Rule

&

Addition Rule

The Multiplication Rule

OR	$\cup$	UNION
AND	$\cap$	INTERSECTION
GIVEN	/	CONDITIONAL

- If events A and B are independent, then the probability of two events, A and B occurring in a sequence or simultaneously is:

$$P(A \text{ and } B) = P(A \cap B) = P(A) \times P(B)$$

- This rule can extend to any number of independent events.

$$P(A \text{ and } B \text{ and } C) = P(A) \times P(B) \times P(C)$$

- ✓ **Sequence** → One event happens **after** another.
- ✓ **Simultaneous** → Events happen **at the same time**.

OR	∪	UNION
AND	∩	INTERSECTION
GIVEN	/	CONDITIONAL

The Multiplication Rule

Example: Events Happening independent & in Sequence

- **Event A:** Toss a coin **(**Getting a Head**)** → P(A) = ½
- **Event B:** Toss the coin again **(**Getting a Head**)** → P(B) = ½
- Since the two tosses are **independent** & in **sequence**

$$P(A \text{ and } B) = P(A) \times P(B) = ½ \times ½ = ¼ = 0.25$$

👉 The probability of getting **two Heads in a row** is **0.25**.



The Multiplication Rule

Example: Events Happening independent & simultaneously

Event A: Getting an even number on Die 1 → P(A) = 3/6 = ½

Event B: Getting a 4 on Die 2 → P(B) = 1/6



Since rolling two dice are **independent** & **simultaneously**

$$P(A \text{ and } B) = P(A) \times P(B) = ½ \times 1/6 = 1/12$$

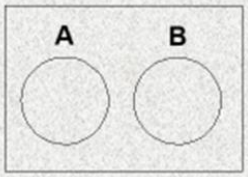
👉 The probability of getting an even number on **Die-1** and 4 on **Die-2** is **1/12**.

Multiplication Rule

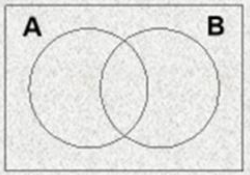
&

Addition Rule

Addition Rule



A and B are M.E.



A and B are not M.E.

A and B are not mutually exclusive

- The probability that at least one of the events A or B will occur, P(A or B), is given by:

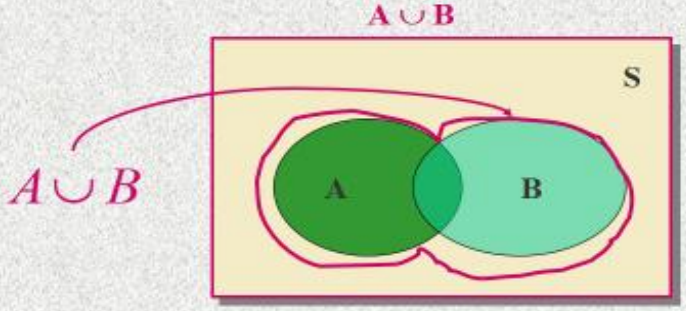
$$P(A\ or\ B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- This formula subtracts the overlap (if both events occur) to avoid double counting.

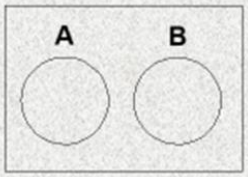
A and B are mutually exclusive

- If events A and B are mutually exclusive, then the addition rule is simplified to:

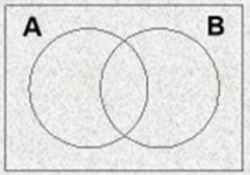
$$P(A\ or\ B) = P(A \cup B) = P(A) + P(B)$$



Addition Rule



A and B are M.E.



A and B are not M.E.

Example: **A and B are not mutually exclusive**

- When two events A and B are **not mutually exclusive,** $P(A \cap B) \neq 0$ and

  $\mathbf{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$

- Suppose that there were 120 students in the classroom, and that they could be classified as follows:

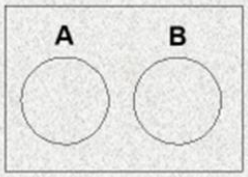
	Brown	Not Brown
Male	20	40
Female	30	30

A: Brown hair
P(A) = 20+30/120 = 50/120

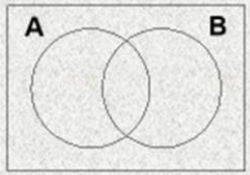
B: Female
P(B) = 30+30/120 = 60/120

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$= 50/120 + 60/120 - 30/120$
$= 80/120$
$= 2/3$

Addition Rule



A and B are M.E.



A and B are not M.E.

Example: **A and B are mutually exclusive**

- When two events A and B are **mutually exclusive,** $P(A \cap B) = 0$ and

  $\mathbf{P(A \cup B) = P(A) + P(B)}$

- Suppose that there were 120 students in the classroom, and that they could be classified as follows:

	Brown	Not Brown
Male	20	40
Female	30	30

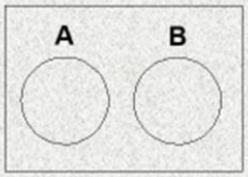
A: male with brown hair
$P(A) = 20/120$

B: female with brown hair
$P(B) = 30/120$

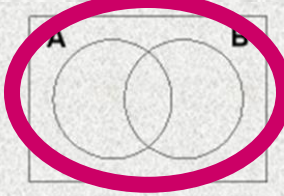
A and B are mutually exclusive, so that

$$P(A \cup B) = P(A) + P(B) = 20/120 + 30/120 = 50/120$$

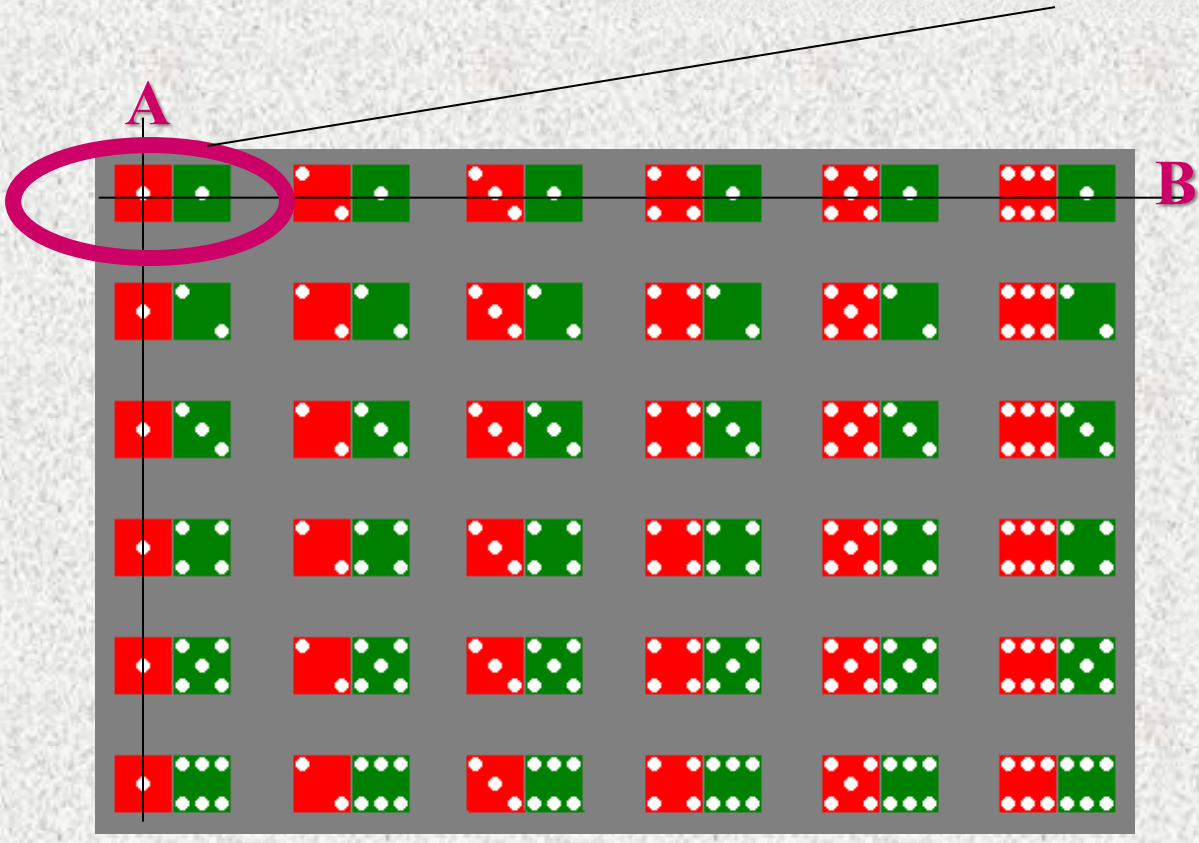
Example: Two Dice



A and B are M.E.



A and B are not M.E.



A: red die show 1

P (A) = 6/36

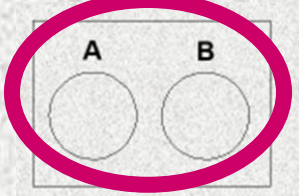
B: green die show 1

P (B) = 6/36

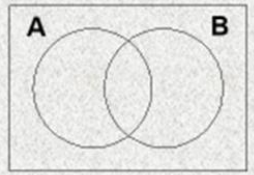
A and B are not mutually exclusive, so that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= 6/36 + 6/36 - 1/36$$
$$= 11/36$$

Example: Two Dice



A and B are M.E.



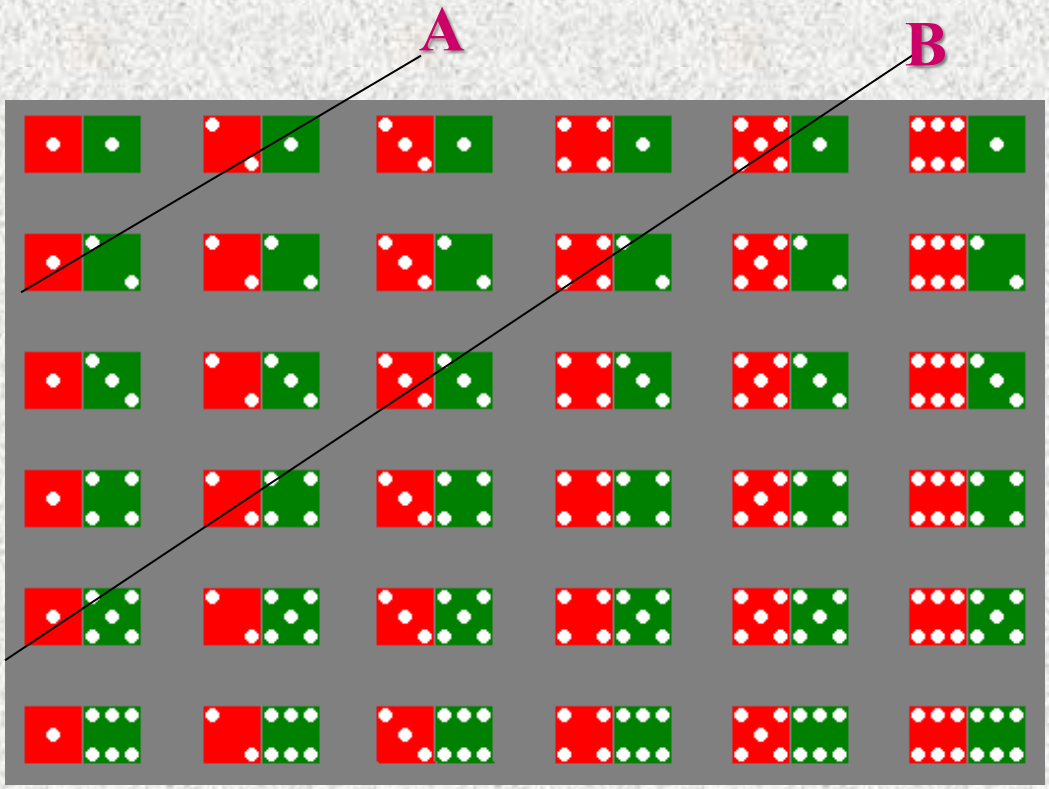
A and B are not M.E.

A: dice add to 3

P (A) = 2/36

B: dice add to 6

P (B) = 5/36



A and B are mutually exclusive, so that

$$P(A \cup B) = P(A) + P(B)$$
$$= 2/36 + 5/36$$
$$= 7/36$$

Conditional Probability

Conditional Probability

- Conditional Probability means the chance of an event happening given that another event has already happened.
- It is written as

  P(B|A)

  (read as *"probability of B given A"*).

- Used when events depend on each other — the first event affects the second.

Conditional Probability

➢ **Formula for Conditional Probability**

👉 Used when you know **A happens first**.

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)}$$

$$P(B \cap A) = P(B|A) \times P(A)$$

👉 Used when you know **B happens first**.

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A|B) \times P(B)$$

Conditional Probability

EXAMPLE 1.

- The following table gives data on the type of car, grouped by petrol consumption, owned by 100 people.

	Low	Medium	High	Total
Male	12	33	7	
Female	23	21	4	
				100

One person is selected at random.

- L is the event "the person owns a low rated car"
- F is the event "a female is chosen".

Find (i) $P(L)$ (ii) $P(F \cap L)$ (iii) $P(F/ L)$

$$P(F \cap L) = P(F|L) \times P(L)$$

Conditional Probability

EXAMPLE 1.

	Low	Medium	High	Total
Male	12	33	7	
Female	23	21	4	
				100

- L is the event "the person owns a low rated car"
- F is the event "a female is chosen".

Find (i) $P(L)$ (ii) $P(F \cap L)$ (iii) $P(F/L)$

(i) P(L)

P(L) = Probability of owning a low-rated car

$$P(L) = \frac{\text{Number of people with low-rated cars}}{\text{Total people}} = \frac{35}{100} = \frac{7}{20}$$

(ii) P(F ∩ L)

P(F ∩ L) = Probability of selecting a **female who owns a low-rated car**

$$P(F \cap L) = \frac{\text{Number of females with low-rated cars}}{\text{Total people}} = \frac{23}{100}$$

Conditional Probability

EXAMPLE 1.

	Low	Medium	High	Total
Male	12	33	7	
Female	23	21	4	
				100

- L is the event "the person owns a low rated car"
- F is the event "a female is chosen".

Find (i) $P(L)$ (ii) $P(F \cap L)$ (iii) $P(F/L)$

(iii) P(F|L)

P(F|L) = Probability of selecting a **female given that the person owns a low-rated car**

$$P(F|L) = \frac{\text{Number of females with low-rated cars}}{\text{Number of people with low-rated cars}} = \frac{23}{35}$$

Conditional Probability

EXAMPLE 1.

	Low	Medium	High	Total
Male	12	33	7	
Female	23	21	4	
				100

- L is the event "the person owns a low rated car"
- F is the event "a female is chosen".

$$P(F \cap L) = P(F|L) \times P(L)$$

$$23/100 = (23/35) \times (7/20)$$

$$23/100 = 161 / 700$$

$$\mathbf{23/100 = 23/100}$$

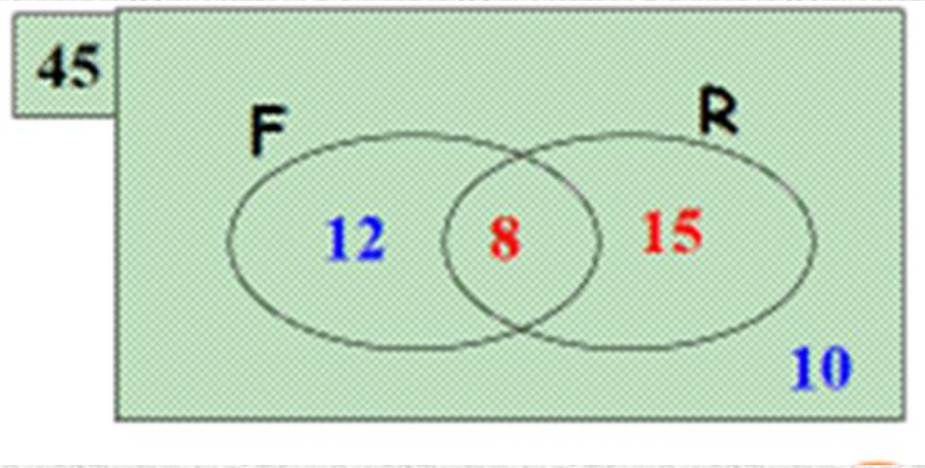
$$\frac{161 \div 7}{700 \div 7} = \frac{23}{100}$$

Conditional Probability

EXAMPLE 2:

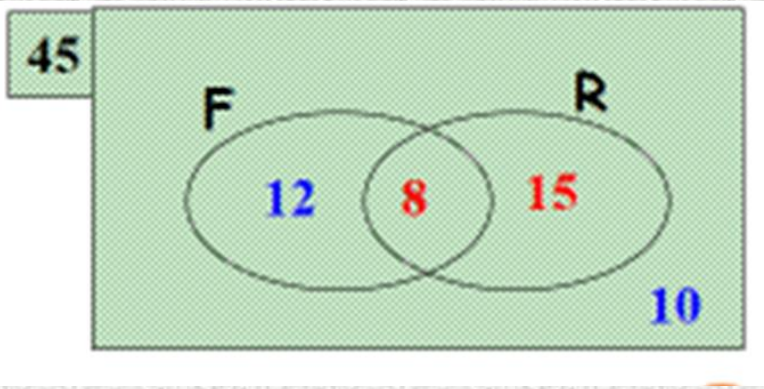
- I have 2 packets of seeds.
  - ✓ One packet has **20** seeds: **8** will give red flowers & **12** blue.
  - ✓ 2nd packet has **25** seeds: **15** will be red and **10** blue.

- Draw a Venn diagram and use it to illustrate the conditional probability formula.



Conditional Probability

EXAMPLE 2:



Let: R be the event " Red flower "

F be the event " First packet "

Find (i) $P(F)$ (ii) $P(R \cap F)$ (iii) $P(R/F)$

(i) P(F)

P(F) = Probability of selecting the First packet

$$P(F) = \frac{\text{Number of seeds in the First packet}}{\text{Total number of seeds}} = \frac{20}{(20+25)} = \frac{20}{45} = \frac{4}{9}$$

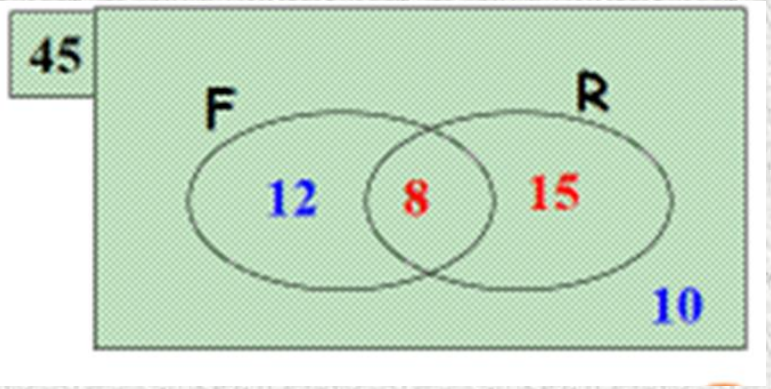
(ii) P(R ∩ F)

P(R ∩ F) = Probability of selecting a red flower seed from the First packet

$$P(R \cap F) = \frac{\text{Number of red seeds in the First packet}}{\text{Total number of seeds}} = \frac{8}{45}$$

Conditional Probability

EXAMPLE 2:



Let: R be the event " Red flower "

F be the event " First packet "

Find (i) $P(F)$ (ii) $P(R \cap F)$ (iii) $P(R/F)$

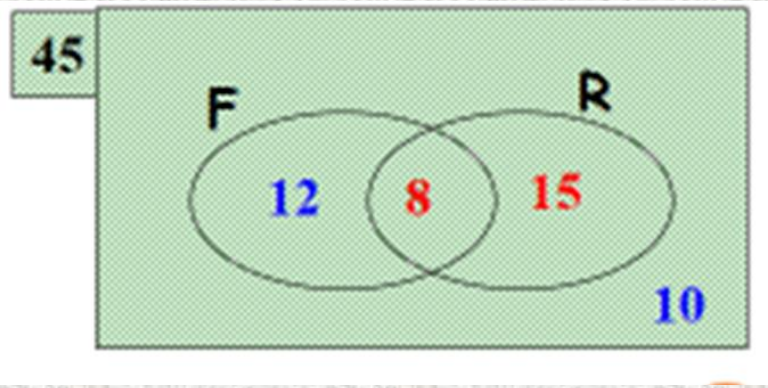
(iii) P(R|F)

P(R|F) = Probability of getting a red flower seed given that the seed is from the First packet

$$P(R|F) = \frac{\text{Number of red seeds in the First packet}}{\text{Number of seeds in the First packet}} = \frac{8}{20} = \frac{2}{5}$$

Conditional Probability

EXAMPLE 2:



Let: R be the event " Red flower "

F be the event " First packet "

$$P(R \cap F) = P(R|F) \times P(F)$$

$$8/45 = (2/5) \times (4/9)$$

$$\mathbf{8/45 = 8/45}$$

THANK YOU